ESP32 Visual Recognition Module

Note: The camera comes from the factory with the factory firmware already flashed. If you have not flashed any other AI demo firmware, there is no need to flash the factory firmware again. The factory firmware of the camera is provided only as a bin file for flashing and does not include the program source code.

1. Connecting the ESP Camera to the Computer

Connect the computer and the ESP32 camera with a Type-C data cable,

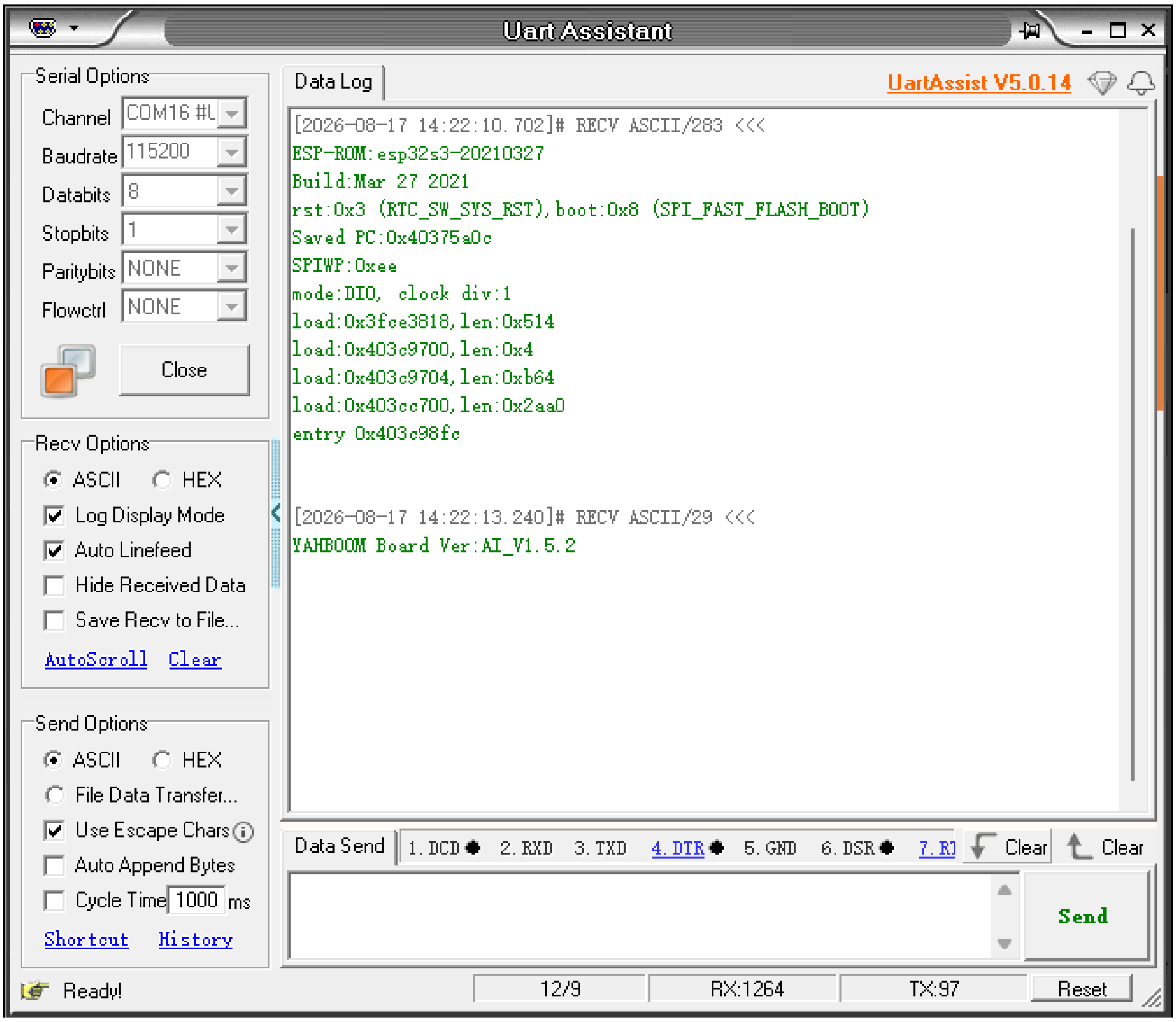
2. Serial Port Configuration

Baud rate 115200, no parity, no hardware flow control, **1** stop bit

If you are using a serial assistant to configure, when sending commands, you should **remove** the auto-append newline (extra bit)

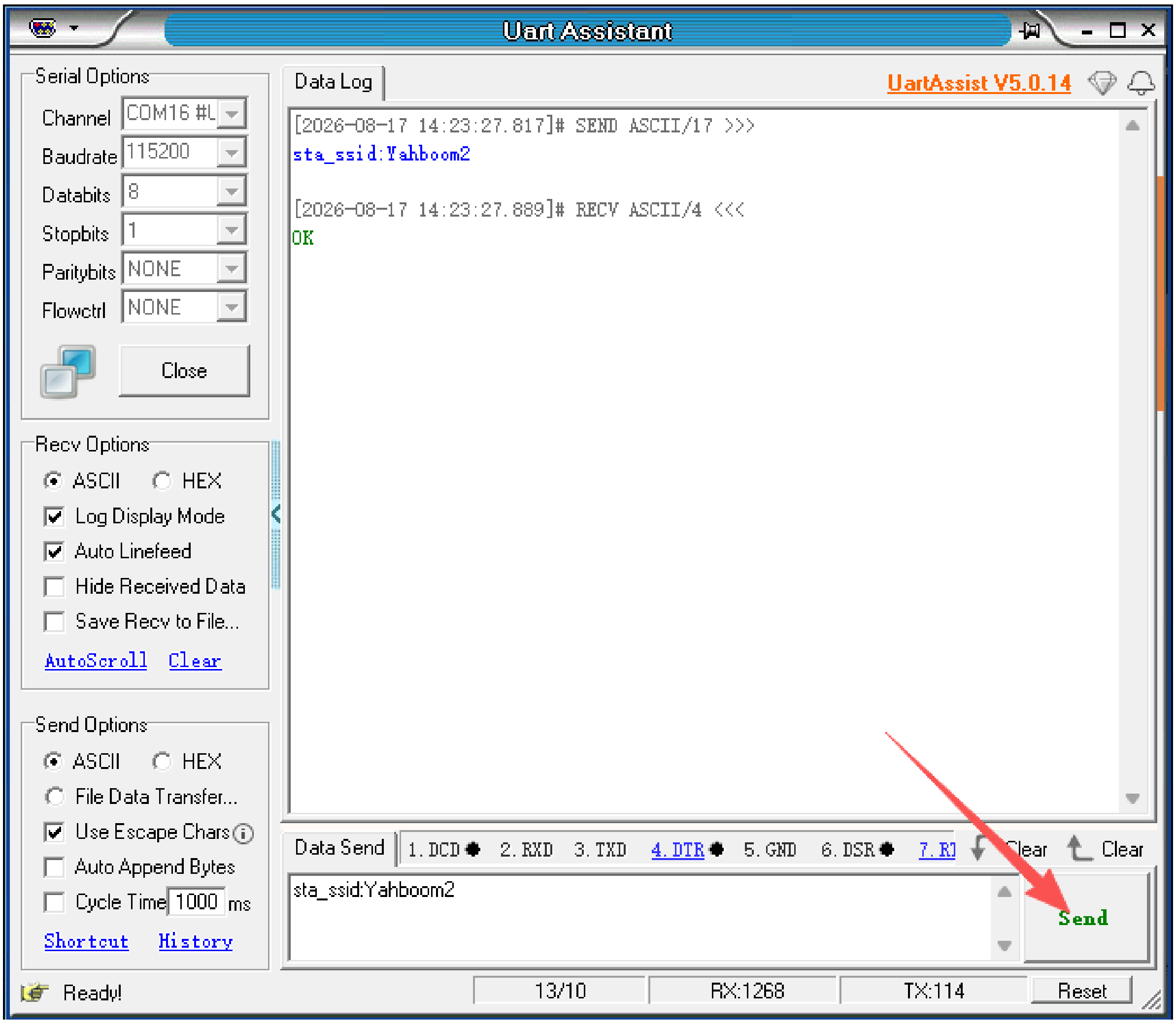
As shown in the figure: the display below indicates that the factory firmware has been flashed. If the following information does not appear, press the RESET button on the module.

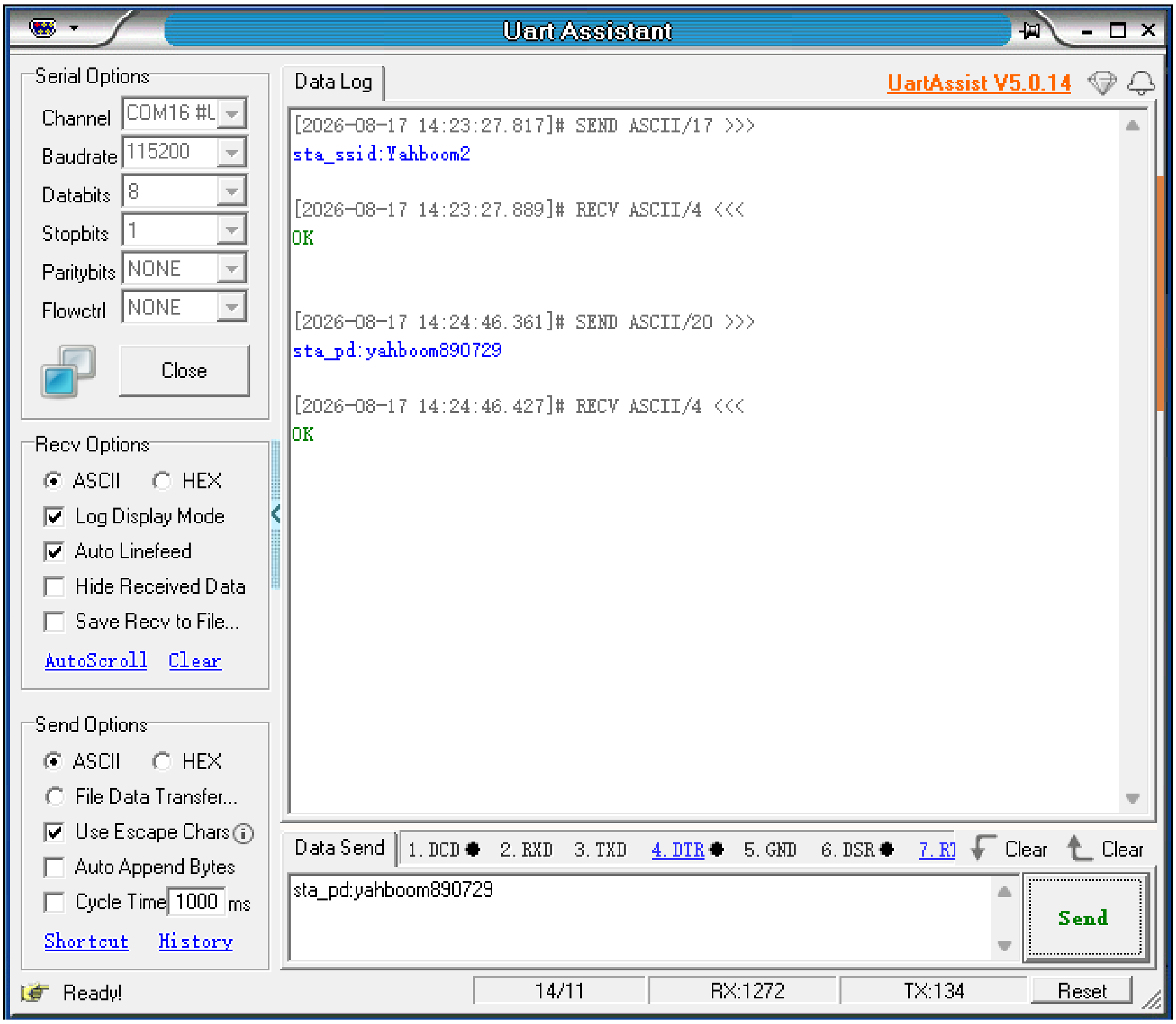
Note: As shown in the figure, version V1.5.2 is the normal mode. If it is V1.5.1, it is the single-demo mode, which does not support all the commands in this tutorial. You can re-flash the V1.5.2 firmware.



2.1 Configuring WiFi Commands in STA Mode (LAN Connection Mode)

Command	Description	Example	Remarks
sta_ssid:	WiFi name to connect to	sta_ssid:yahboom	yahboom: the WiFi to connect to
sta_pd:	WiFi password to connect to	sta_pd:12345678	12345678: the password of the WiFi to connect to


Uart Assistant
Serial Options
Channel COM16 #L
Baudrate 115200
Databits 8
Stopbits 1
Paritybits NONE
Flowctrl NONE
Close
Recv Options
ASCII HEX
Log Display Mode
Auto Linefeed
Hide Received Data
Save Recv to File...
AutoScroll Clear
Send Options
ASCII HEX
File Data Transfer...
Use Escape Chars
Auto Append Bytes
Cycle Time 1000 ms
Shortcut History
Data Log
UartAssist V5.0.14
[2026-08-17 14:23:27.817]# SEND ASCII/17 >>>
sta_ssid:Yahboom2
[2026-08-17 14:23:27.889]# RECV ASCII/4 <<<
OK
Data Send
1. DCD 2. RXD 3. TXD 4. DTR 5. GND 6. DSR 7. RI
Clear Clear
sta_ssid:Yahboom2
Send
Ready!
13/10
RX:1268
TX:114
Reset


Uart Assistant
Serial Options
Channel COM16 #L
Baudrate 115200
Databits 8
Stopbits 1
Paritybits NONE
Flowctrl NONE
Close
Recv Options
ASCII HEX
Log Display Mode
Auto Linefeed
Hide Received Data
Save Recv to File...
AutoScroll Clear
Send Options
ASCII HEX
File Data Transfer...
Use Escape Chars
Auto Append Bytes
Cycle Time 1000 ms
Shortcut History
Data Log
UartAssist V5.0.14
[2026-08-17 14:23:27.817]# SEND ASCII/17 >>>
sta_ssid:Yahboom2
[2026-08-17 14:23:27.889]# RECV ASCII/4 <<<
OK
[2026-08-17 14:24:46.361]# SEND ASCII/20 >>>
sta_pd:yahboom890729
[2026-08-17 14:24:46.427]# RECV ASCII/4 <<<
OK
Data Send
1. DCD 2. RXD 3. TXD 4. DTR 5. GND 6. DSR 7. RI
Clear Clear
sta_pd:yahboom890729
Send
Ready!
14/11
RX:1272
TX:134
Reset

Notes

1. Whether it is sta_ssid or sta_pd, it must be followed by an English punctuation mark, such as **comma (,), colon (:), period (.), etc.**
2. When the WiFi to connect to has no password, send **sta_pd:** once like this.
3. The above commands will return **OK** on success. If **OK is not returned**, check:
   (1) Whether the module is in single-demo mode (YAHBOOM Board Ver:AI_V1.5.1). Version V1.5.1 is single-demo mode and has no network configuration function; you need to flash the factory firmware (V1.5.2 version)
   (2) Check the serial port wiring
4. When sending the **sta_ssid:** command like this, it will return the string **fail,ssid is null!**, indicating that the WiFi name to connect to cannot be empty
5. Commands can be sent entirely in uppercase or entirely in lowercase
6. The WiFi name and password cannot exceed 30 characters in length, otherwise the configuration fails
7. The WiFi name and password cannot contain Chinese characters
8. **Every time the WiFi password is changed, the module resets automatically. If only the WiFi name is changed, you need to manually power-cycle the module.**

2.2 Configuring WiFi Commands in AP Mode (Module Self-Hosted Hotspot Mode)

Command	Description	Example	Remarks
ap_ssid:	WiFi name to set	ap_ssid:my_wifi	my_wifi: the WiFi to set
ap_pd:	WiFi password to set	ap_pd:12345678	12345678: the password of the WiFi to set

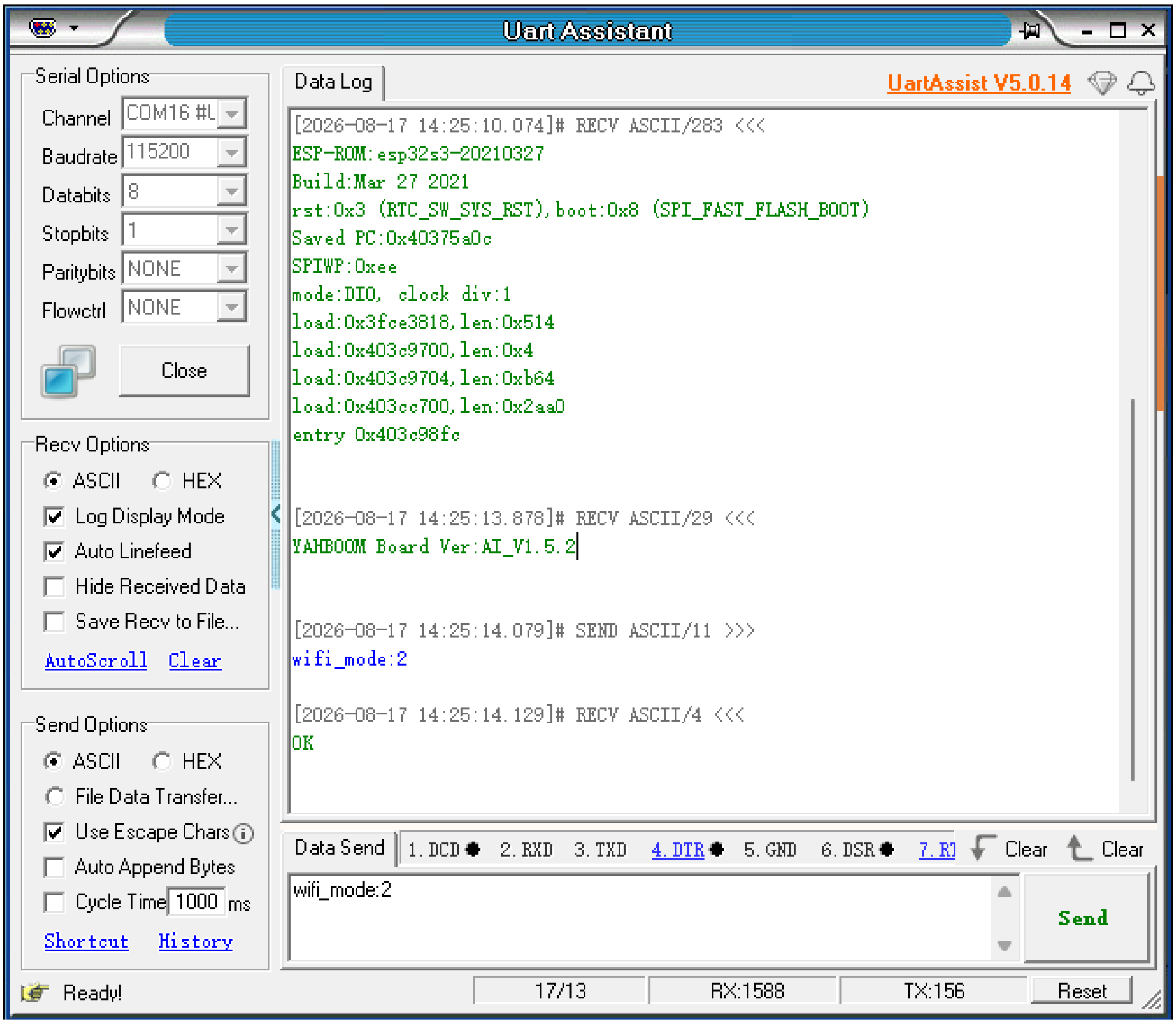
You can use the default hotspot name. The password can be left unset.

Notes

1. Whether it is ap_ssid or ap_pd, it must be followed by an English punctuation mark, such as **comma (,), colon (:), period (.), etc.**
2. When the WiFi to set has no password, send **ap_pd:** once like this.
3. The above commands will return **OK** on success. If **OK is not returned**, check:
   (1) Whether the module is in single-demo mode (YAHBOOM Board Ver:AI_V1.5.1). Version V1.5.1 is single-demo mode and has no network configuration function; you need to flash the factory firmware (V1.5.2 version)
   (2) Check the serial port wiring
4. When sending the **ap_ssid:** command like this, it will return the string **fail,AP_Name is null!**, indicating that the WiFi name to set cannot be empty
5. Commands can be sent entirely in uppercase or entirely in lowercase
6. The WiFi name and password cannot exceed 30 characters in length, otherwise the configuration fails
7. The WiFi name and password cannot contain Chinese characters
8. **Every time the WiFi password is changed, the module resets automatically. If only the WiFi name is changed, you need to manually power-cycle the module.**

2.3 Configuring the WiFi Mode

Command	Description	Example	Remarks
wifi_mode:	Configure the WiFi mode	wifi_mode:2	0: AP mode 1: STA mode 2: AP+STA mode

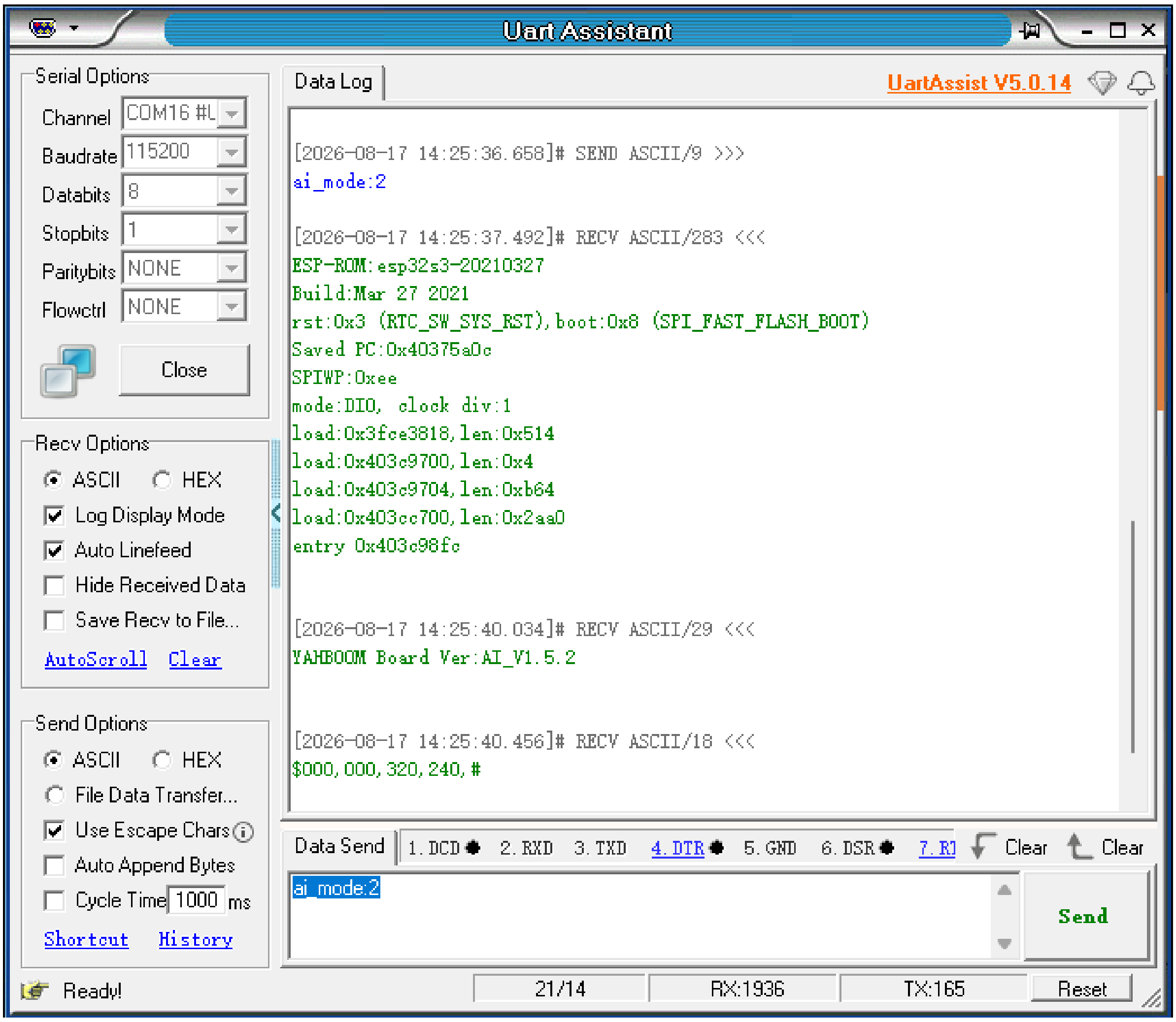


Notes

1. wifi_mode must be followed by an English punctuation mark, such as **comma (,), colon (:), period (.), etc.**
2. wifi_mode can only be set to 3 modes. If a negative number is passed, it defaults to AP mode; if a number greater than 2 is passed, it defaults to AP+STA mode
3. The above command will return **OK** on success. If **OK is not returned**, check:
   (1) Whether the module is in single-demo mode (YAHBOOM Board Ver:AI_V1.5.1). Version V1.5.1 is single-demo mode and has no network configuration function; you need to flash the factory firmware (V1.5.2 version)
   (2) Check the serial port wiring

2.4 Setting the AI Mode

Command	Description	Example	Remarks
ai_mode:	Configure the AI recognition mode	ai_mode:0	0: Normal mode 1: Cat/dog detection 2: Face detection 3: Color recognition 4: Face recognition 5: QR code recognition



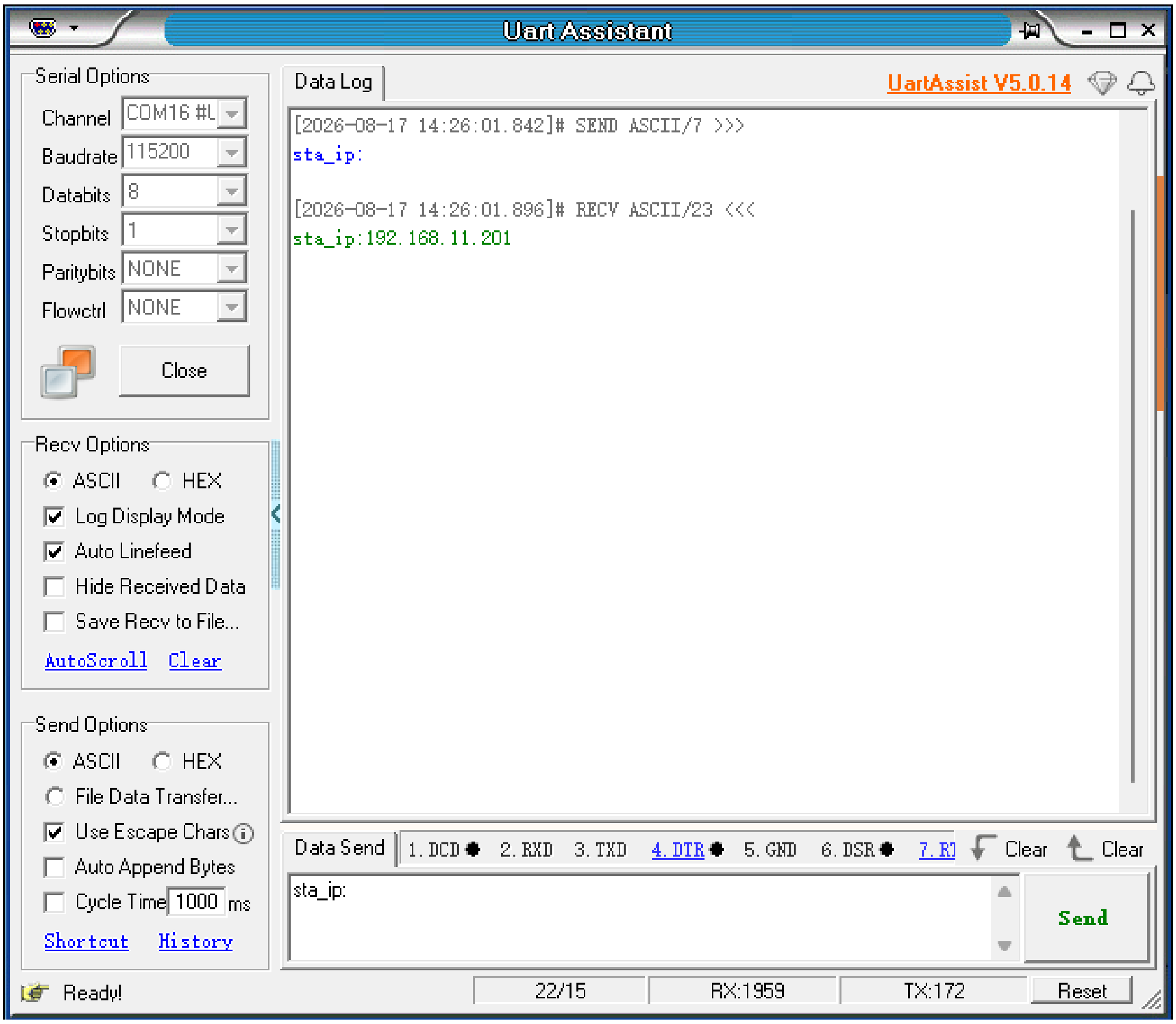
After modifying the ai_mode, the module resets.

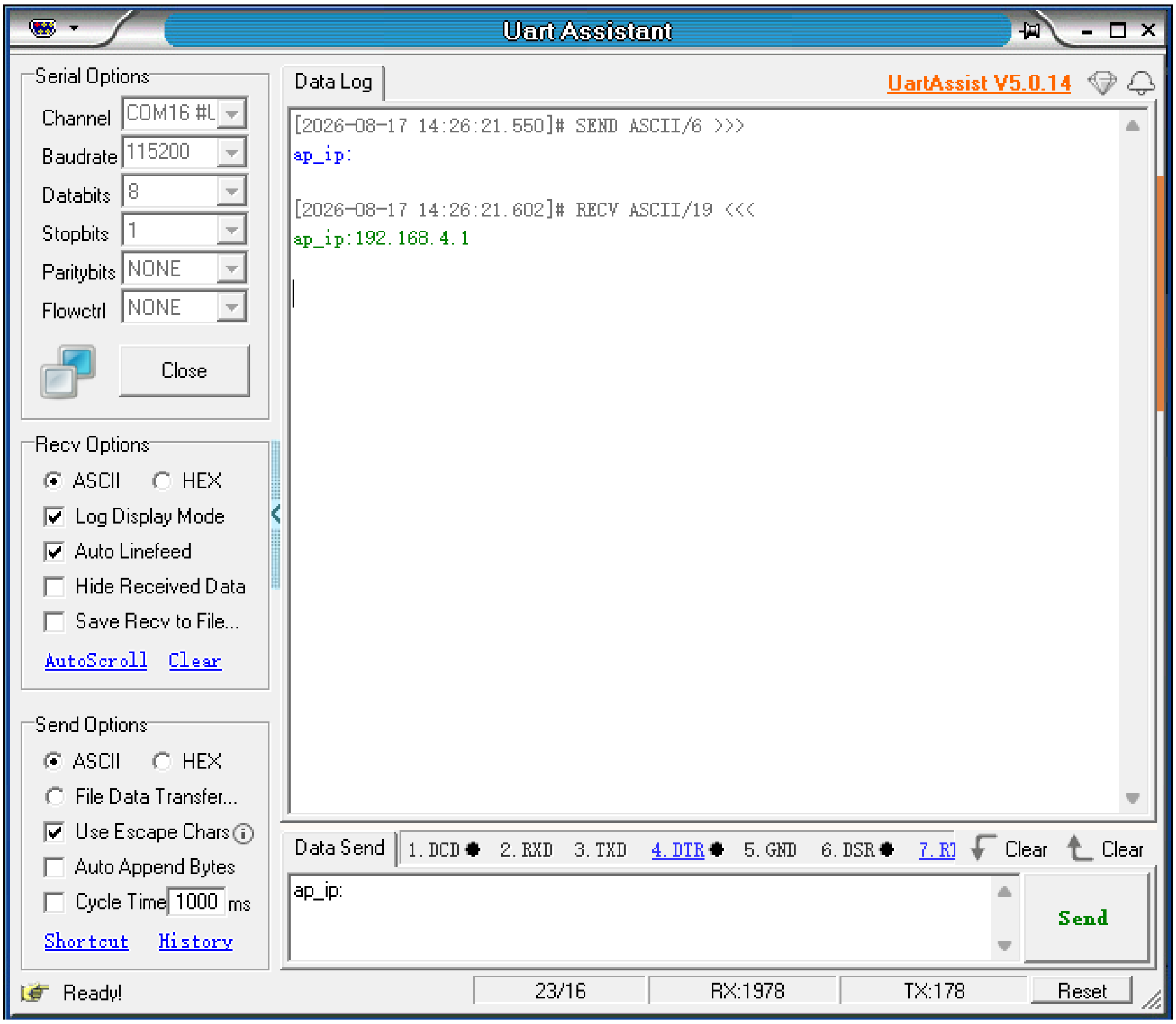
Notes

1. ai_mode must be followed by an English punctuation mark, such as **comma (,), colon (:), period (.), etc.**
2. ai_mode can only be set to 5 modes. If an illegal number is passed, it defaults to the normal mode.
3. Every time the mode is changed, the image transmission module restarts automatically. After the module restarts successfully, **you need to refresh the webpage or re-enter the phone app**
4. ai_mode: can also be sent in all-uppercase English as AI_MODE:
5. Factory firmware mode: the default is the normal mode.
6. In face recognition mode, the Type-C serial port configuration and network transparent transmission functions become unavailable, mainly because there is not enough space to create the tasks. If you need these functions, do not use face recognition mode; face detection mode works normally.

2.5 Reading the IP Address

sta_ip	**Query the sta-mode IP address**	**sta_ip**	**Returns the IP address connected to the LAN (e.g., sta_ip:192.168.2.111)**
ap_ip	Query the ap-mode IP address	ap_ip	Returns the IP address of the module's own WiFi hotspot (e.g., ap_ip:192.168.4.1)

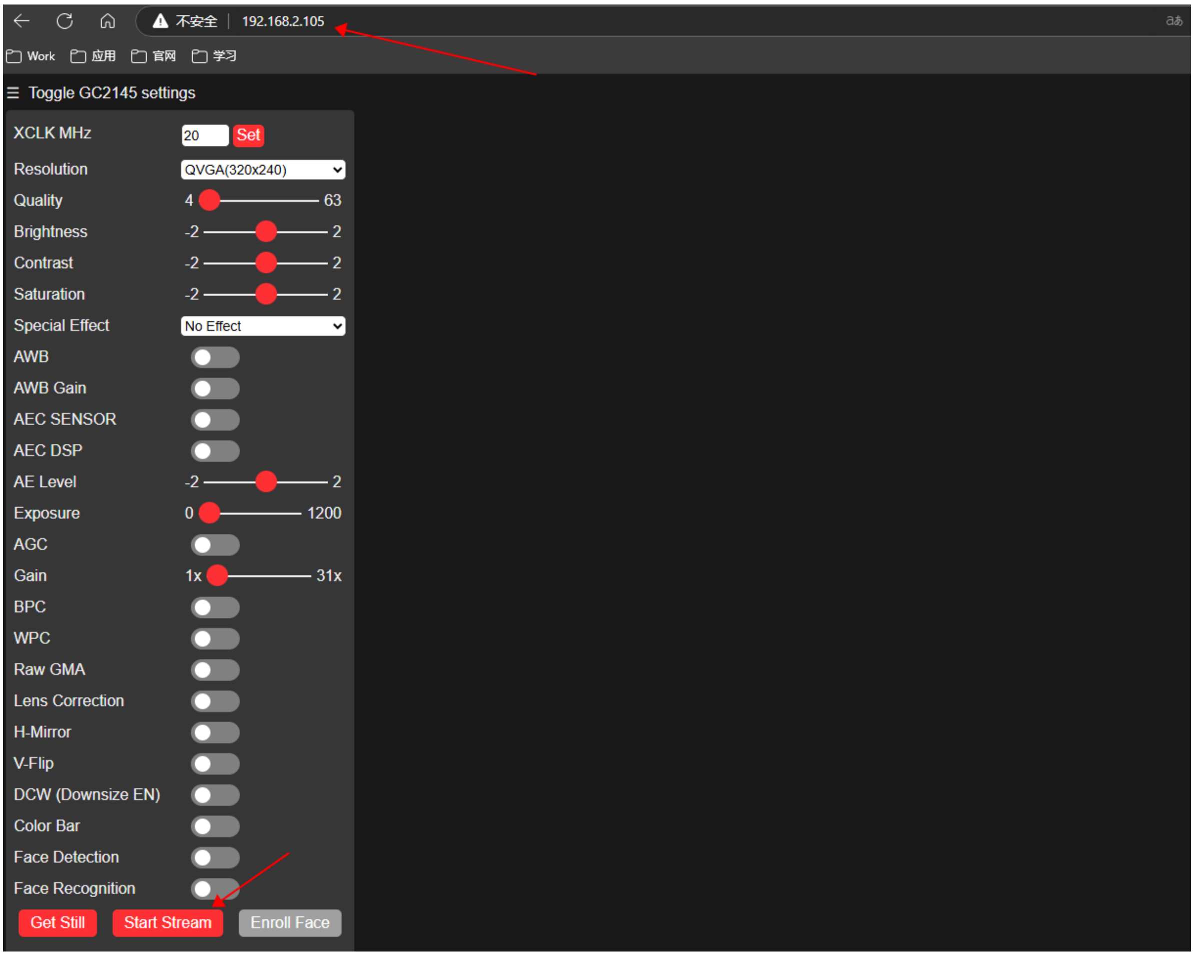




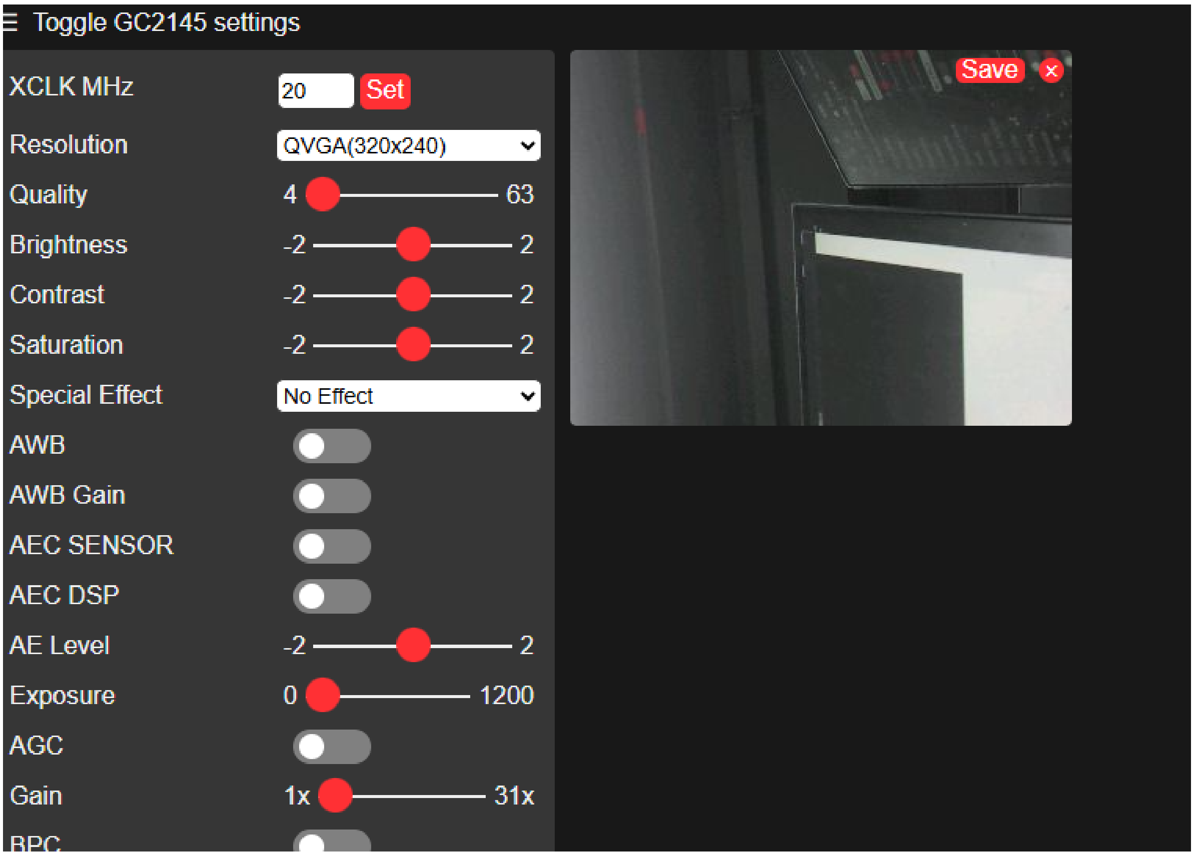
3. Viewing the Image in a Browser

3.1 Remote Viewing over the LAN

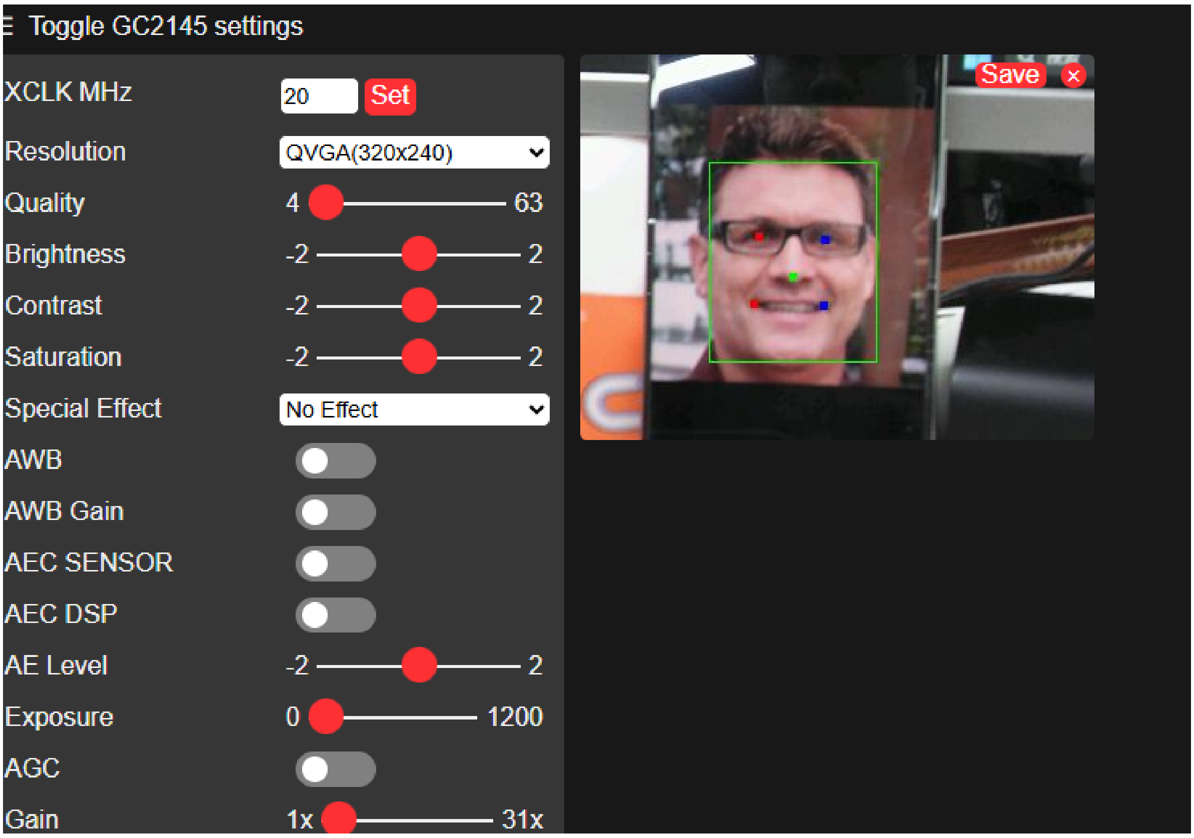
Now that we know the sta_ip (192.168.2.105) obtained by the module when it connects to the LAN, we can use this IP to view the real-time image on the network,



Click start stream to view the image,

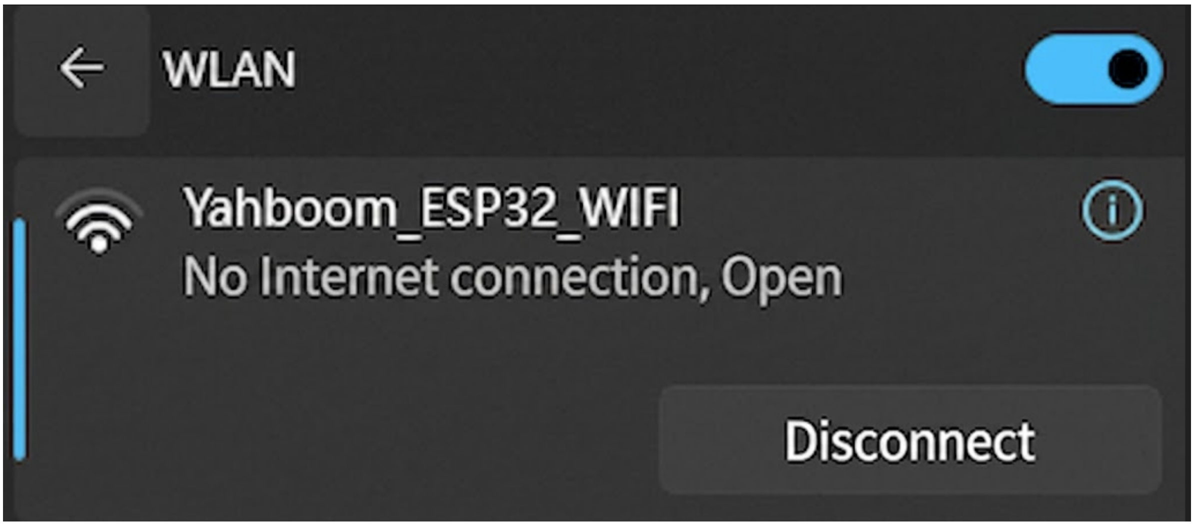


Because the face detection mode (ai_mode:2) was just set, placing a face in front of the camera can be recognized normally,

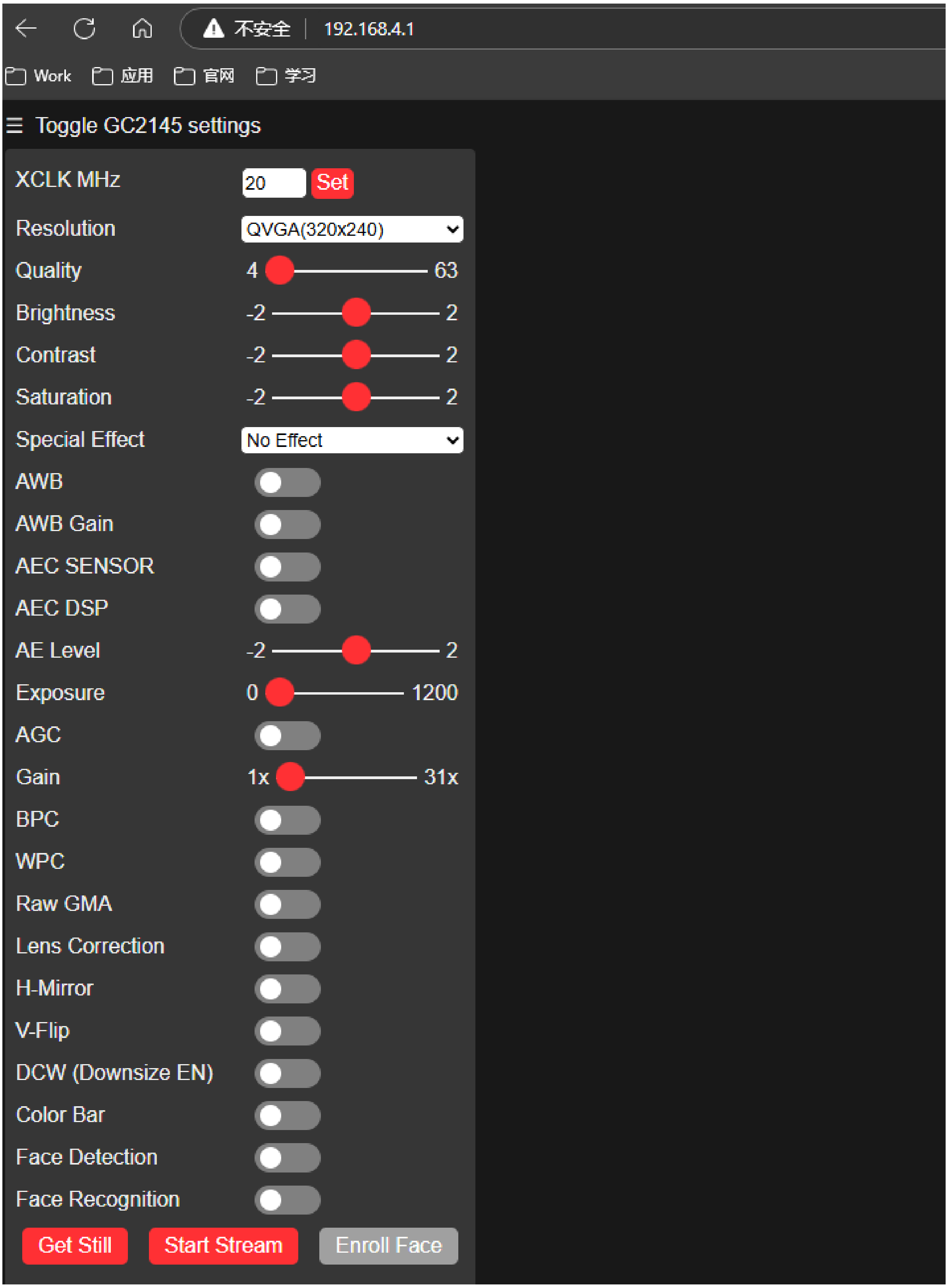


3.2 Remote Viewing via the Module Hotspot

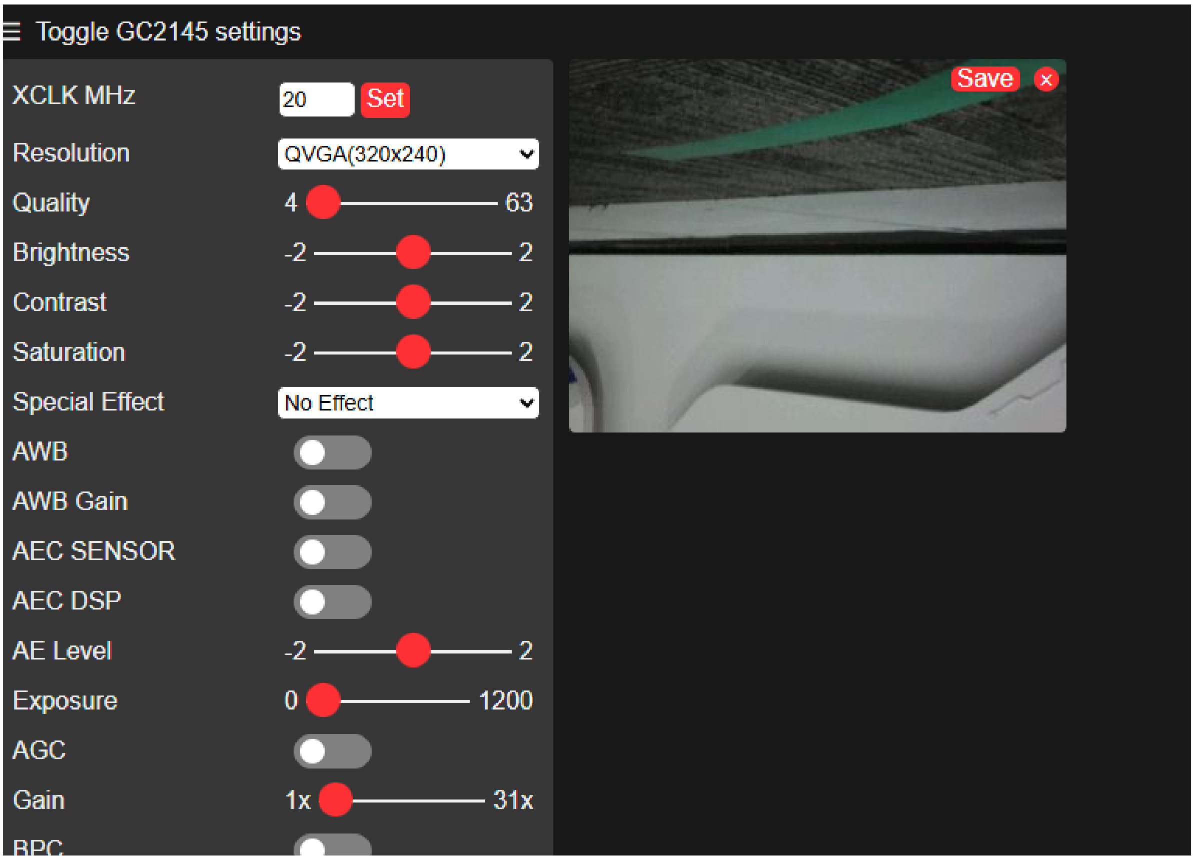
Connect the computer to the module's hotspot



Open a browser and enter the IP address 192.168.4.1. This address is the fixed address of the hotspot,



Click start stream to view the image,



Because the face detection mode (ai_mode:2) was just set, placing a face in front of the camera can be recognized normally, **(other AI modes can be configured by yourself)**